

Digital Systems Design Lab

(BECE102P)

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ModelSim Software Download Link

<https://tinyurl.com/mvucv8yh>

The rubrics for lab as follows:

- ✓ Assessment 1 – 10 marks
- ✓ Assessment 2– 10 marks
- ✓ Assessment 3– 10 marks
- ✓ **Assessment 4– 10 marks**
- ✓ Assessment 5 – 10 marks
- ✓ Assessment 6 – 10 marks
- ✓ **FAT – 50 marks**

Your notebook must be clearly labelled on the cover with the following information:

Top of Form

Course Name : Digital Systems Design Laboratory

Name :

Register no :

Slot/Batch timing :

Lab faculty Name :

Theory Faculty Name :

Lab Record must contain

- Aim
- Tool Required
- Theory
- Logic Diagram
- Procedure
- Verilog Model
- Output { Screen Shot} (Result)
- Inference

Experiment

Write the Verilog HDL code for all the logic gates and verify the truth table using Model sim.

Logic Gates

- **Logic gates** are electronic circuits that operate on one or more input signals to produce an output signal
- *Gate inputs* are driven by voltages having two nominal values, e.g. 0 V and 5 V representing logic 0 and logic 1
- The *output* of a gate provides two nominal values of voltage 0 V and 5 V representing logic 0 and logic 1
- The **truth table** of a logic system (e.g. digital electronic circuit) describes the output(s) of the system for given input(s).
- In other words, a truth table is a table of all possible combinations of the variables, showing the relation between the values that the variables may take and the result of the operation.

Types of Logic gates

1) NOT/INVERTOR

2) AND

3) OR

4) BUFFER

5) XOR, XNOR

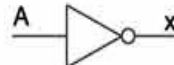
6) NAND

7) NOR

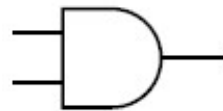
Basic logic gates

Universal logic gates (gates from which any digital circuit can be built)

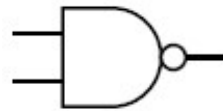
Logic Gates

Name	NOT	AND	NAND	OR	NOR	XOR	XNOR																																																																																																
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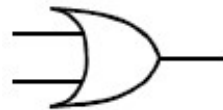
Verilog Primitive gates



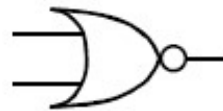
and(out, in1, in2)



nand(out, in1, in2)



or(out, in1, in2)



nor(out, in1, in2)



xor(out, in1, in2)



xnor(out, in1, in2)

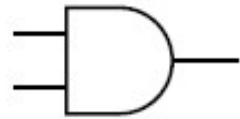


buf(out, in)

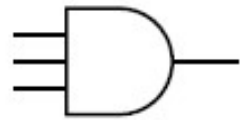


not(out, in)

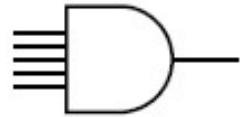
Verilog Gates (more than two inputs)



`and(out, in1, in2)`



`and(out, in1, in2, in3)`



`and(out, in1, in2, in3, in4, in5)`

Verilog HDL code

```
module logic_gates(Y_not,Y_and,Y_or,Y_buf,Y_nand,Y_nor,Y_xor,Y_xnor,A,B);
  input A,B;
  output Y_not,Y_and,Y_or,Y_buf,Y_nand,Y_nor,Y_xor,Y_xnor;

  // not gate
  not g0(Y_not,A);

  //and gate
  and g1(Y_and,A,B);

  //or gate
  or g2(Y_or,A,B);

  //buffer gate
  buf g3(Y_buf,A);

  //nand gate
  nand g4(Y_nand,A,B);

  //nor gate
  nor g5(Y_nor,A,B);

  //xor gate
  xor g6(Y_xor,A,B);

  //xnor gate
  xnor g7(Y_xnor,A,B);

endmodule
```

